



W I S P

Household Monitoring AI — Project Briefing v4

AI plans. AI monitors. Humans execute.

SHIELDS HOUSEHOLD · WES / RIA / ROMY / WHIDBEY
MYSTIC, CT · JULY 2026 · EXECUTIVE SUMMARY + FLEET ARCHITECTURE

EXECUTIVE SUMMARY

EXSUM — Wisp v1

BLUF: Wisp assumes 100% of household Planning (approval-gated) and Monitoring (autonomous) for 72 audited responsibilities, while its Execution share is locked at exactly 0% by database constraint. Target effect by day-90 re-audit: Ria's monitoring load falls from 56% to under 20% and her planning load from 56% to under 30% — the mental load moves to software without the family surrendering a single decision or action.

72

RESPONSIBILITIES

56%→<20%

RIA MONITORING

0%

WISP EXECUTION —
LOCKED

\$50/mo

ENDURING CEILING

Situation. A Fair Play-style audit of 72 household responsibilities shows the invisible work — noticing, tracking, re-planning — concentrated on one adult: Ria owns 40 of 72 Planning rows and 40 of 72 Monitoring rows, with daily-cadence monitoring at 11 items to 1. Execution is already near parity. The problem is not who does the work; it is who has to notice it.

Mission. Deploy a self-hosted household AI that plans and monitors so the humans only execute, measured by a formal day-90 re-audit against the original spreadsheet.

Execution. Three phases plus one hardware gate: Foundation (schema, seed, calendar, heartbeat), Capture + Accountability (Telegram, extraction, one-tap confirmations, escalation for critical/pact items only), a staged fleet-validation gate on a single tablet, then Surfaces (SMS, wall dashboards, room-to-room intercom, announcements). Budget is two ledgers: \$50/month enduring ceiling (projected \$25–40) and a one-time \$500–750 fleet capex, gated.

Risk posture. No cameras, ever. No always-on microphones. A child's schedule stays off speakers and public surfaces. All ingested text is treated as untrusted data. Escalation exists only where a human opted in.

PLAIN LANGUAGE

How Wisp Works

Wisp watches the calendar, the pantry patterns, the school email, and whatever the family tells it in passing — a voice note, a text, a tap on the kitchen display. It turns all of that into three simple things: a morning briefing, a small set of "this needs a human today" signals, and a proposed weekly plan that a person approves or edits. When someone does the thing, they tap **Done** once. That's the entire ceremony.

What Wisp never does: the task itself, or deciding a task is finished. It cannot mark work complete — only Wes or Ria can, with one tap. It never nags: routine items surface once and then quietly become input for next week's plan.

THE CONTRACT

AI plans. AI monitors. Humans execute.

The execution owner field in the database is a human-only list — there is no "wisp" value to assign. Any code path that would let the AI complete a task is defined as a bug. Plans are proposals until a human approves. Monitoring runs on its own.

Soft accountability — Option A (locked)

- **One tap is the only execution event.** Dashboard, phone, or SMS keyword.
- **Photos are optional state reports** — never required, never AI-reviewed, never a gate.
- **Escalation (push → SMS → call) exists only for critical signals and pact items**, and cancels the instant a confirmation lands — transactionally.
- **Pacts are self-opt-in only.** You can promise your own week; nobody pacts on your behalf.
- **Routine items never nag.** Missed ones roll silently into the weekly pattern.
- **Announcements are event-only.** Wisp never editorializes over the speakers.

BASELINE

The Audit Numbers

DIMENSION	RIA	WES	SHARED	OTHER	DAY-90 TARGET
Planning (72 rows)	40 (56%)	15	16	1 tbd	Ria < 30% · Wisp proposes all
Monitoring (72 rows)	40 (56%)	10	21	1 tbd	Ria < 20% · Wisp autonomous
Execution (72 rows)	27	24	19	1 outsource · 1 tbd	Unchanged · Wisp = 0% exactly
Daily-cadence monitoring	11 items	1 item	—	—	→ Wisp

Seed encoding: every planning and monitoring owner flips to *wisp*; the original human is preserved in *owner_legacy* so the day-90 re-audit compares like-for-like. The generator script asserts these totals on every regeneration — the seed cannot silently drift.

Monitoring tiers

TIER	ROWS	MECHANISM	EXAMPLES
T1 calendar	49	Deterministic date math on the 15-minute heartbeat; injected into the Wisp Google Calendar	Appointments, laundry cycles, school deadlines, renewals
T2 state	8	Consumption / threshold models with capture assist	Groceries, dog food & meds, household supplies, trash
T3 human capture	15	Humans see, Wisp routes — capture → observation → signal, no nagging	Tidying standards, meal quality, repairs, hosting

No cameras. Ever. The novel core is the routing layer: observations map to responsibilities, not to people.

Architecture & Stack

Application

- Next.js 15 + React 19 + TypeScript strict + Tailwind 4
- Neon Postgres with row-level security (Drizzle *crudPolicy*; owner role reserved for migrations)
- Vercel: hosting, 15-min heartbeat cron, daily/weekly agents (AI SDK v5)
- Claude API: extraction, morning briefing, weekly planner — never in the heartbeat
- MCP: Google Calendar (Wisp calendar only) · Gmail read-only · Mem (Wisp collection)

Channels & capture

- Telegram bot (week 2, \$0) — capture, confirmations, briefings
- Twilio SMS (week 3+; A2P 10DLC paperwork starts week 2)
- Voice — Phase 2 gate via ConversationRelay + Claude
- Adapters: telegram · sms · email · pwa · cozyla touch · intercom · fieldy/plaud (pending API verification)
- STT: OpenAI gpt-4o-mini-transcribe — intercom push-to-talk only

Security posture

- All ingested content is untrusted data, never instructions (prompt-injection / lethal-trifecta discipline; delimited data blocks in every extraction prompt).
- Human confirmation before any outbound side effect; signed-webhook verification on every inbound path; secrets in env only.
- Child-data rule: when in doubt about any surface — wall, speaker, notification — don't.

HARDWARE

Display Fleet, Intercom & Announce

DEVICE	ROLE	QTY	EST. COST	NOTES
Cozyla display (owned)	Kitchen anchor	1	\$0	Wisp calendar via native sync day one; kiosk sideload is a Phase 2.5 test
Galaxy Tab A9+ 11"	Anchor rooms	1 → 2-3	\$160-220 ea	Staged buy: one unit must pass the gate before the rest
Galaxy Tab A9 8.7"	Small rooms	0 → 2-3	\$110-140 ea	Post-gate only
Fully Kiosk Browser Plus	Kiosk + REST control	per device	~\$11 one-time	Port 2323, LAN/Tailscale only — never WAN-exposed
Home Assistant (Green)	Future sensor brain	optional	\$0-199	Post-v1; fleet code stays HA-independent
Tailscale / CF Tunnel	Remote control plane	—	\$0	Fleet REST reachable only on the tailnet

Phase 2.5 — the fleet gate

One A9+ is purchased first and must pass five checks — kiosk lockdown + autostart, mic capture for push-to-talk, remote TTS over Tailscale, dashboard compact mode, Cozyla parity — before any further hardware spend. Failure routes to a Lenovo Tab M11 evaluation, not a bigger order.

Intercom v1 — push to talk

Hold the orb on any tablet → speak → release. Audio is transcribed (gpt-4o-mini-transcribe), spoken into the selected rooms via the kiosk TTS API, and both parents receive the transcript by SMS. The audio blob is deleted the moment transcription completes. No live audio, no wake words, no always-on mics.

Announce — two inputs only

Wisp event templates, and SMS from the two verified parent numbers. Enforced server-side: 280-character cap, URLs stripped, quiet hours 20:30-08:00 ET (queued, not dropped), event-only content. Romy's location patterns never reach a speaker.

One dashboard component tree serves every screen: full grid at ≥1000px, compact single-column below. No per-device forks.

PLAN

Phases & Gates

PHASE	SCOPE	HARD GATE
1 — Foundation week 1	Drizzle schema + RLS (incl. confirmation, device, criticality, pact), seed load, two-user auth, Wisp calendar, T1 engine v0, heartbeat cron	Seed verifies (72 rows, counts match); T1 dates visible on the Cozyla; heartbeat green 3 consecutive days
2 — Capture + accountability weeks 2-3	Telegram, extraction + dead-letter, one-tap confirm endpoints, escalation engine (critical + pact, cancel-on-confirm), daily briefing, T2 thresholds, school email, weekly planner. A2P paperwork starts top of week 2	Phone → capture → briefing untouched by hand; Ria approves a plan from her phone; a critical escalation cancels on confirm
2.5 — Fleet gate	Buy ONE A9+; validate kiosk, mic, remote TTS, compact mode, Cozyla parity	All five pass → fleet purchase approved; any fail → M11 fallback eval first
3 — Surfaces week 4+	SMS live, production dashboard v4, fleet rollout, intercom v1, announce pipeline, voice go/no-go	Both adults capture daily for a week; zero missed T1 items; one announcement lands in every room + SMS transcript
Day-90 — Re-audit	Rerun the audit spreadsheet, blank scoring table	Ria monitoring <20% · planning <30% · Wisp execution = 0% exactly. Misses trigger scope review

Budget — Two Ledgers, Never Mixed

Enduring — ceiling \$50/mo (projected \$25-40)

ITEM	MONTHLY
Neon Postgres	\$0-19
Vercel	\$0-20
Claude API (extraction, briefing, planner)	~\$5-15
Twilio number + SMS	~\$2-4
STT (intercom PTT only)	~\$1-3
Tailscale · ntfy · Fully Kiosk REST	\$0

One-time capex — approved \$500-750

ITEM	ONCE
Tab A9+ (staged first unit)	\$160-220
Remaining satellites (post-gate)	\$330-500
Fully Kiosk Plus licenses	~\$11 × devices
HA Green (optional, post-v1)	\$199

Escape valves: reduce satellite count; A9 instead of A9+ in small rooms; Telegram stays the permanent primary channel if A2P approval stalls past four weeks; voice ships only if SMS adoption proves out. Spend grows only against measured time savings.

Open questions

- Fieldy / Plaud API surfaces — verify before building adapters
- Cozyla kiosk sideload feasibility ("Cozyla Frames" community first)
- Voice worker host (Fly vs Railway vs Render) — Phase 2 decision
- Primary approval surface: PWA vs wall display vs ntfy buttons
- "Date night" row — a household conversation, not code